

CS231n Deep Learning for Computer Vision

Course Website

Assignment 2

This assignment is due on **Wednesday, May 07 2025** at 11:59pm PST.

Starter code containing Colab notebooks can be [downloaded here](#).

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Setup

Please familiarize yourself with the [recommended workflow](#) before starting the assignment. You should also watch the Colab walkthrough tutorial below.



CS231N Google Colab Assignment Workflow Tutorial

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Watch on

Note. Ensure you are periodically saving your notebook (**File -> Save**) so that you don't lose your progress if you step away from the assignment and the Colab VM disconnects.

While we don't officially support local development, we've added a **requirements.txt** file that you can use to setup a virtual env.

Once you have completed all Colab notebooks **except** `collect_submission.ipynb`, proceed to the [submission instructions](#).

Goals

In this assignment you will practice writing backpropagation code, and training Neural Networks and Convolutional Neural Networks. The goals of this assignment are as follows:

- Implement **Batch Normalization** and **Layer Normalization** for training deep networks.
- Implement **Dropout** to regularize networks.
- Understand the architecture of **Convolutional Neural Networks** and get practice with training them.
- Gain experience with a major deep learning framework, **PyTorch**.
- Understand and implement RNN networks. Combine them with CNN networks for image captioning.

Q1: Batch Normalization

In notebook `BatchNormalization.ipynb` you will implement batch normalization, and use it to train deep fully connected networks.

Q2: Dropout

The notebook `Dropout.ipynb` will help you implement dropout and explore its effects on model generalization.

Q3: Convolutional Neural Networks

In the notebook `ConvolutionalNetworks.ipynb` you will implement several new layers that are commonly used in convolutional networks.

Q4: PyTorch on CIFAR-10

For this part, you will be working with PyTorch, a popular and powerful deep learning framework.

Open up `PyTorch.ipynb`. There, you will learn how the framework works, culminating in training a convolutional network of your own design on CIFAR-10 to get the best performance you can.

Q5: Image Captioning with Vanilla RNNs

The notebook `RNN_Captioning_pytorch.ipynb` will walk you through the implementation of vanilla recurrent neural networks and apply them to image captioning on COCO.

Submitting your work

Important. Please make sure that the submitted notebooks have been run and the cell outputs are visible.

Once you have completed all notebooks and filled out the necessary code, you need to follow the below instructions to submit your work:

1. Open `collect_submission.ipynb` in Colab and execute the notebook cells.

This notebook/script will:

- Generate a zip file of your code (`.py` and `.ipynb`) called `a2_code_submission.zip`.
- Convert all notebooks into a single PDF file.

If your submission for this step was successful, you should see the following display message:

```
###      Done!      Please      submit      a2_code_submission.zip      and  
a2_inline_submission.pdf to Gradescope. ###
```

Note: When you have completed all notebooks, please ensure your most recent kernel execution order is chronological as this can otherwise cause issues for the Gradescope autograder. If this isn't the case, you should restart your kernel for that notebook and rerun all cells in the notebook using the Runtime Menu option "Restart and Run All".

2. Submit the PDF and the zip file to [Gradescope](#).

Remember to download `a2_code_submission.zip` and `a2_inline_submission.pdf` locally before submitting to Gradescope.

